

Experimental Assessment of Non-Contact Sensing Technologies for Remote River Level Monitoring

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Abstract

River water level is a key variable for disaster mitigation, agriculture, environmental monitoring, and the planning of hydroelectric projects. But the rivers that most need monitoring are usually remote, lack cellular coverage, and expose contact sensors to debris and strong currents that corrode or damage them over time, making reliable, timely measurement difficult. This project addresses this by building a non-contact, low-power monitoring system, testing three sensing technologies (ultrasonic, radar, and LiDAR) in the lab, at sea, and on a river under different mounting angles and conditions, with readings sent end to end over LoRaWAN and cellular backhaul to a live visualization platform. The ultrasonic sensor was the cheapest but needed a clear line of sight, the radar sensor was the most consistently accurate but failed sharply once tilted past its rated angle, and the LiDAR sensor reached the longest range but was highly sensitive to water clarity, sunlight, and surface conditions. An adaptive transmission strategy was also built in, raising the reporting rate during rapid water-level changes and lowering it during stable flow to save power without losing responsiveness during floods. These findings offer practical guidance for choosing the right sensor for a given deployment site.

 **Code:** <https://github.com/joudoq/Experimental-Assessment-of-Non-Contact-Sensing-Technologies-for-Remote-River-Level-Monitoring>

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1 Introduction

1.1 Motivation and Real-World Impact

River water level is a key variable for disaster mitigation, agriculture, and environmental monitoring, and it strongly affects the planning and operation of hydroelectric plants. However, for this data to actually be useful, it needs to reach the decision makers quickly. Since the rivers that need to be monitored are often in remote areas, we need a complete system that can measure water levels without physically touching the water to protect the equipment from debris and floods and that can transmit that data wirelessly, even in areas where there is no cellular service.

1.2 Problem Statement

Contact sensors are exposed to debris and subject to damage by strong currents. Over time they can clog, corrode, and drift out of calibration, and during floods they may be damaged. A non-contact sensor mounted above the water avoids these problems and can last longer.

The deployment sites add further constraints. Because they often are remote and hard to reach, the solution must be low in cost, run for long periods on limited power, and need little maintenance. A device that must be serviced or replaced often is not practical.

Within these constraints, two questions remain. First, which non-contact technology gives the most accurate and reliable readings across different conditions. Second, how to transmit these readings reliably from the riverbank to a live dashboard in places where cellular service is not available.

1.3 Proposed System

The proposed system measures water level without contact and delivers the data end to end, from the riverbank to a visualization platform. Three sensors are compared under the same conditions: ultrasonic, radar, and a LiDAR. Each sensor is evaluated for accuracy against a tape measure reference. The readings are transmitted over LoRaWAN, chosen for its long range and low power. Data from the LoRaWAN gateway is transmitted by cellular to The Things Network (TTN) Network Server.

2 Related Work

Non-contact water level sensing has been studied from several angles: sensing accuracy, cost, and connectivity in remote settings.

Paul, Buytaert, and Sah [1] evaluated terrestrial LiDAR as a technique for measuring river water levels, and report that it can reach sub-centimeter precision under favorable conditions, while also documenting how factors such as beam incidence angle, surface turbulence, and range affect the quality of the return signal. This is directly relevant to interpreting the LiDAR results in this project, particularly the water-penetration behaviour observed under the calm-water conditions tested.

Kabi et al. [2] present a low-cost river water-level acquisition system built around an ultrasonic sensor and a LoRa radio link, aimed at regions without reliable cellular coverage. Their system validates the combination of an inexpensive non-contact sensor with a long-range, low-power wireless link as a practical architecture for remote monitoring, which mirrors the end-to-end architecture (sensor + LoRaWAN/cellular + visualization) targeted by this project.

Martinez-Osuna et al. [3] survey low-cost water level measurement technologies – including ultrasonic and radar options – for coastal and citizen-science deployments, and compare them on

cost, ease of deployment, and maintenance. Their comparison criteria (cost, robustness, ease of installation) complement the accuracy- and repeatability-focused comparison performed in this project, and will be useful when the final sensor recommendation is made.

Taken together, these works confirm that (i) LiDAR can offer the highest precision but is more sensitive to environmental conditions and cost, (ii) ultrasonic sensors paired with LoRa are an established low-cost combination for remote deployments, and (iii) a systematic side-by-side comparison of ultrasonic, radar, and LiDAR under the same test conditions, which is the gap this project addresses, has not been covered in the literature we have examined.

3 Methodology

3.1 System Overview

The system evaluates three different non-contact sensing technologies: ultrasonic, radar, and LiDAR. Each sensor node reports its readings over LoRaWAN, carrying real-time data wirelessly from the riverbank. Each node is also built to be cost-efficient, so it can stay in the field for years with minimal maintenance. Figure 1 shows this end-to-end data path, from the sensing node to the visualization platform.

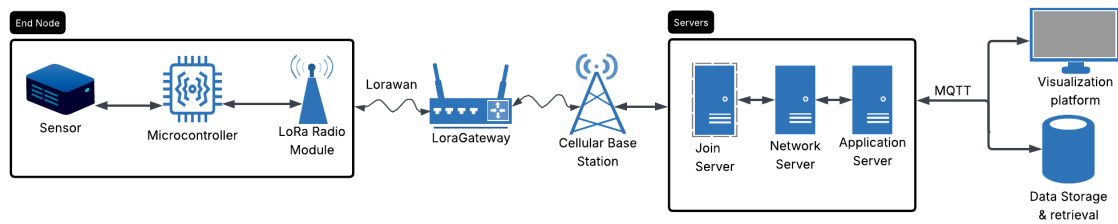


Figure 1 End-to-end data path: the end node (sensor, microcontroller, and LoRa radio module) transmits over LoRaWAN to a LoRa gateway; readings then reach the servers (join, network, and application servers), either directly or via a cellular base station, and are finally delivered over MQTT to the visualization platform and to data storage.

Network Infrastructure: Sensor uplinks received by a LoRaWAN gateway are transmitted using a cellular modem to The Things Network (TTN) LoRaWAN Network Server (LNS), as shown in Figure 1. Each end device is registered to a dedicated TTN application, and for the LiDAR sensor a JavaScript payload formatter decodes the base64-encoded uplinks into distance readings. A Python script reads and decodes the latest message from each device, and writes each record (distance, timestamp, SNR, and RSSI) into an InfluxDB time-series database. The stored readings are then visualized through a Grafana dashboard connected to InfluxDB.

According to the LoRaWAN protocol, downlink packets can also be transmitted from the LNS through the cellular network to the gateway and from it to the end device. This can be used, for instance, to change the transmitting interval, or for firmware updates.

3.2 System Implementation

All three sensors estimate the distance d to the water surface from the time t it takes for an emitted signal traveling at a known propagation speed v to reach the receiver:

$$d = \frac{v \cdot t}{2}$$

where v is the speed of sound for the ultrasonic sensor and the speed of light for the radar and LiDAR sensors.

Notice that the speed of sound depends on the temperature of the medium, whereas that of the electromagnetic wave is independent of temperature. The propagation speed of a 40 kHz ultrasonic wave in fresh water at 24 °C is about 1494 m/s, versus 345 m/s in air; an ultrasonic wave will penetrate several kilometers into water. For light, the propagation speed is about 300 000 km/s in air and about 226 258 km/s in fresh water. The penetration depth of a 905 nm wave in fresh water is approximately 14.3 cm, while that of a 60 GHz wave is only 0.3 mm.

The three sensors evaluated in this study are summarized in Table 1 and described below.

Sensor	Type	Wave Type/Freq.	Range	Beam Width
Milesight EM500-UDL	Ultrasonic	Acoustic (40 kHz)	0.25–10 m	~ 60°
Milesight EM410-RDL	Radar	EM (60 GHz)	0.3–12 m	8°
Garmin LiDAR-Lite v3HP	LiDAR	EM (331 THz)	up to 40 m	~ 0.5°

Table 1 Sensors used in this study.

- **Ultrasonic (Milesight EM500-UDL):** Ultrasonic distance and level sensor. Commercial off-the-shelf device with integrated microcontroller and LoRaWAN transceiver.
- **Radar (Milesight EM410-RDL):** Radar distance sensor. Commercial off-the-shelf device with integrated microcontroller and LoRaWAN transceiver.
- **LiDAR:** a custom unit specifically assembled and programmed for this project, consisting of:
 - **Ranging sensor:** a Garmin LiDAR-Lite v3HP.
 - **Microcontroller:** an Arduino Diecimila.
 - **Radio module:** a Seeed Studio LoRa-E5 (Wio-E5) transceiver.
 - **Case:** a 3D-printed case shaped like the LiDAR's lens (Figure 2).
 - **Power consumption:** 0.49 W (5.12 V, 96 mA).

You can view the firmware that runs the custom LiDAR node [here](#).

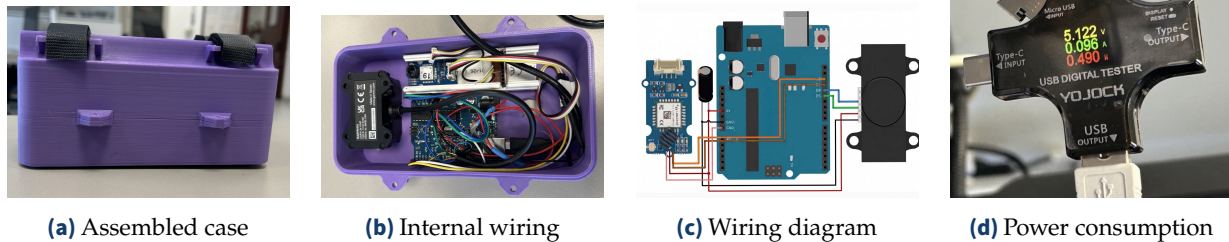


Figure 2 The custom LiDAR node.

The principle of lidar ranging relies on the rugosity of the reflective surface to generate nonspecular reflection (i.e., scattering) of the incipient laser beam [1]. The major difference with terrestrial lidar applications is the lower reflectivity of water surfaces for infrared radiation compared to solid objects. A calm water surface may behave as a purely specular surface, which produces reflections instead of dispersing incipient energy. This means that the reflected beams often miss the receiver altogether [1]. In practice, however, small turbulence or particles present in the water will allow some reflection even in calm water.

Adaptive Transmission Strategy: For the LiDAR node, the firmware dynamically adjusts the transmission interval based on river activity (Table 2), maximizing the sleep interval when the water level is stable and increasing the transmission frequency during rapid change. This reduces power consumption and operational cost in remote LoRaWAN deployments. It was implemented only for the LiDAR node, since it is the only sensor with fully custom, in-house firmware; the two off-the-shelf Milesight units run closed vendor firmware that does not expose this level of control, but the transmission interval can be modified by means of downlink messages from the LoRaWAN Network Server.

Condition	Change Between Readings	Transmission Interval
Stable flow	< 1% of distance	15 minutes
Moderate activity	1%–3.5% of distance	5 minutes
Rapid change	> 3.5% of distance	1 minute

Table 2 Adaptive transmission interval strategy.

4 Results

4.1 Lab Experiments

Measurement results of the ultrasonic, radar, and LiDAR sensors under several test conditions were compared against manual tape measurements. The trials were conducted at different locations inside the laboratory.

4.1.1 Ultrasonic Sensor – Lab Trials

Test	Condition	Actual (m)	Sensor (m)
1	Sensor placed horizontally at the middle of a table facing the wall	3.83	1.33
2	Sensor placed vertically above a container with 5 cm of water (<i>actual distance measured to the water surface, not the container floor</i>)	0.52	0.52
3	Sensor placed vertically above a container with 5 cm of water from a farther distance (<i>actual distance measured to the water surface, not the container floor</i>)	0.93	0.94
4	Sensor placed horizontally at the table edge facing the wall	2.12	2.120

Table 3 Ultrasonic sensor: lab trials.

Test 1 gave an incorrect reading because the table surface interfered with the beam. In Tests 2 and 3, the readings confirmed that the wave was correctly reflected from the surface of the water. Test 4, at the table edge, matched the tape-measured distance and shows that the wide 60° beam can be intercepted by any surface that falls inside its cone.

Test	Condition	Actual (m)	Sensor (m)
1	Sensor placed horizontally at the edge of a table facing the wall	0.66	0.661
2	Sensor placed horizontally at the middle of the table facing the wall	0.88	0.882
3	Sensor placed horizontally facing the wall from a longer distance	3.90	3.899
4	Sensor placed vertically facing a water container in the lab (<i>actual distance measured to the water surface, not the container floor</i>)	0.79	0.789
5	Sensor placed horizontally facing a long narrow corridor	12.5	4.285

Table 4 Radar sensor: lab trials.

4.1.2 Radar Sensor – Lab Trials

Test 2 gave an accurate reading despite the small table interference, because the radar beamwidth is much narrower than the ultrasonic sensor; in Test 4, the radar sensor successfully reflected off the calm, clear water surface. Test 5 gave incorrect measurements because the sidewalls interfered with the beam.

4.1.3 LiDAR Sensor – Lab Trials

We then tested the LiDAR sensor the same way (Table 5). Over calm, clear water, unlike the radar sensor, it did not reflect off the water surface; instead, it penetrated the water and reflected off the bottom of the container. The reported distance was higher because light travels at about 2.26×10^8 m/s in water, instead of 3×10^8 m/s in air, so the extra time the beam spent underwater was misread as extra distance.

Test	Condition	Actual (cm)	Sensor (cm)
1	Sensor placed vertically on the table, facing the floor	75	73
2	Sensor placed vertically at a higher position, facing the floor	200	202
3	Sensor placed vertically facing the floor, from a farther distance	445	449
4	Sensor placed horizontally, facing the wall	100	108
5	Sensor placed horizontally facing 5 cm of water in a container	70	83

Table 5 LiDAR sensor: lab trials.

Test 4 shows a larger deviation from the actual distance than the other solid-surface tests, possibly pointing to a systematic error at that orientation, and Test 5 was an incorrect measurement because the sensor did not reflect off the water surface, consistent with the water-penetration behaviour described above.

4.2 Sea Experiments

To evaluate sensor performance, a field experiment was done at the Trieste seashores (Figure 3).

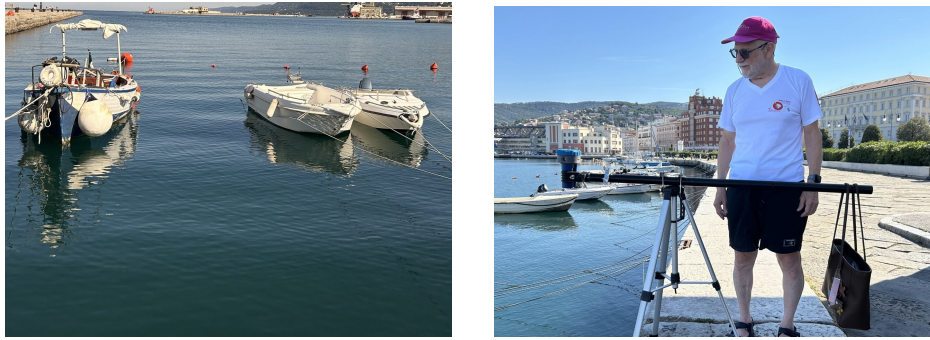


Figure 3 Calm-sea test site.

4.2.1 Ultrasonic Sensor – Sea Trial

Sea trials revealed that the ultrasonic sensor's wide 60° acoustic beam requires careful installation. As shown in Figure 4(a), mounting the sensor too close to the pier caused the beam to intersect the concrete wall before reaching the water. This structural interference resulted in a false echo and an incorrect short reading (0.905 m). When the sensor was repositioned further outward (Figure 4(b)), the measurement was accurate (2.266 m). Therefore, mounting brackets must be designed long enough to prevent the sensor's beam from hitting adjacent structures.



Figure 4 Ultrasonic beam blocked by the pier (a) vs. clear of it (b).

Test	Condition	Actual (m)	Sensor (m)
1	Sensor placed vertically close to the pier edge, facing calm sea water (Figure 4a)	2.25	0.905
2	Sensor placed vertically away from the pier edge, facing calm sea water (Figure 4b)	2.25	2.266

Table 6 Ultrasonic sensor: sea trial.

4.2.2 Radar Sensor – Sea Trial

Test	Condition	Actual (m)	Sensor (m)
1	Sensor placed vertically facing the floor	0.96	0.964
2	Sensor placed close to the pier edge facing calm sea water (Figure 5)	2.48	2.468

Table 7 Radar sensor: sea trial.

Test 2 was an accurate reading, because the radar beamwidth is much narrower than the ultrasonic sensor's, so it was not affected by the pier edge the way the ultrasonic sensor was in Test 1 above.

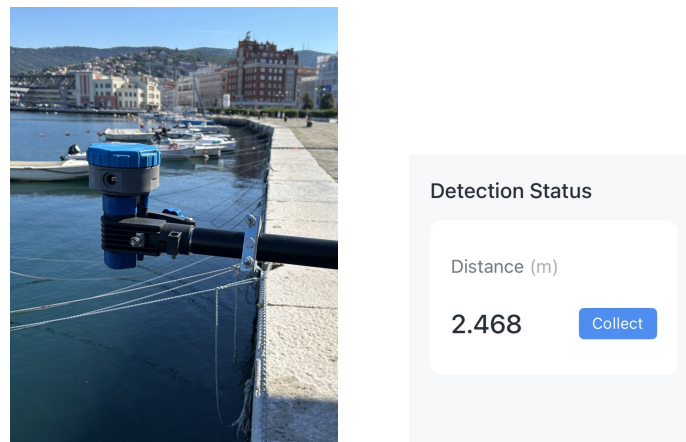


Figure 5 Radar sensor at the pier edge.

4.3 River Experiments

To evaluate sensors' performance, a field experiment was done at the Timavo River, located in the Province of Trieste in north-eastern Italy. During the test, the air temperature was 29 °C and the water temperature was 12 °C.

The three sensors were mounted on a horizontal bar above the river (Figure 6). The ultrasonic sensor was positioned first, since it has the widest beam and is therefore the most sensitive to placement, followed by the radar sensor, and finally the LiDAR sensor. The bar was tilted to a series of fixed angles, and four consecutive readings were recorded from each sensor at every angle. We also intentionally disabled the radar sensor's built-in 15° tilt limit for this test, to see how it behaves once pushed past its rated operating range.

In the river trial, the ground truth could not simply be read off a tape measure, since tilting the bar continuously changes each sensor's height above the water. So we measured the fixed arm length of each sensor along the bar ($L = 84$ cm for the ultrasonic, $L = 61$ cm for the radar, and $L = 40$ cm for the LiDAR sensor) and combined that with simple trigonometry to calculate the expected distance to the water at each tilt angle. Because each sensor sits at a different point along the bar, this gives each of them a slightly different expected distance.

Table 8 summarizes, for each sensor and angle, the mean of the four repeated readings and the percentage error against this calculated ground truth. Figure 7 plots the same mean distances as a bar chart, and Figure 8 shows the raw variation of the three sensors, illustrating just how much noisier the LiDAR reading is even when its mean is accurate.



Figure 6 Overview of the test setup at the Timavo River.

Angle	Ultrasonic		Radar		LiDAR	
	Mean (m)	Error	Mean (m)	Error	Mean (m)	Error
0°	2.566	−2.07%	2.617	−0.13%	2.692	+2.77%
10°	2.658	−3.92%	2.664	−2.25%	2.785	+3.57%
15°	2.825	−0.45%	2.757	−0.76%	—	—
20°	2.886	−0.74%	0.947	−66.51%	2.680	−2.79%
25°	2.974	−0.03%	0.985	−65.78%	2.740	−1.76%
30°	3.033	−0.23%	—	—	—	—

Table 8 Sensor accuracy at each tilt angle (river trial). Each value is the mean of four repeated readings taken from every sensor at every angle.

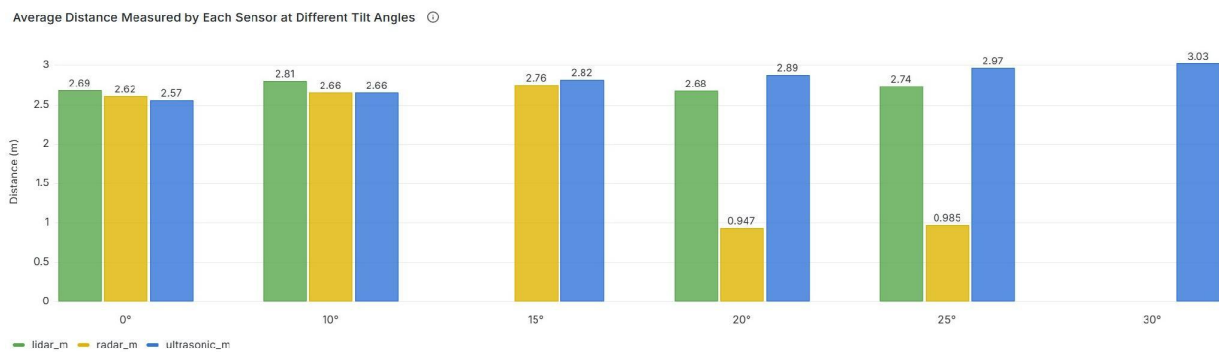


Figure 7 Average distance measured by each sensor at different tilt angles.

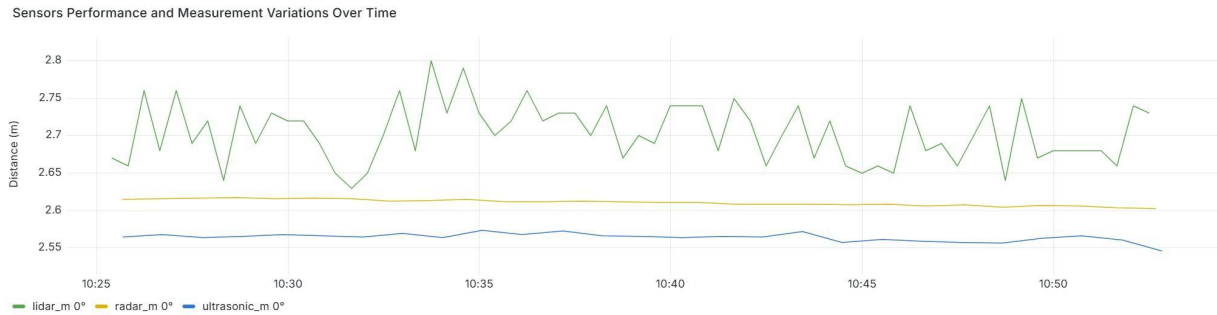


Figure 8 Sensors performance and measurement variations over time.

A second field visit to the Timavo River was carried out to collect additional data beyond the tilt-angle trial reported above. The complete dataset and results details of all the practical experiments is available in the project's GitHub repository.

5 Discussion

The lab experiments showed that the wide 60° beam of the ultrasonic sensor is not a problem on its own; it only fails when objects block its path, but it works perfectly when placed in an open space. Radar and LiDAR sensors, which have narrow beams, easily avoided physical obstacles in these tight spaces. However, while narrow beams help avoid physical structures, they interact differently with the water surface. For instance, the LiDAR showed conflicting results between the lab and the river. In the lab it penetrated clear water rather than reflecting off the surface, leading to incorrect distance readings. However, during the Timavo River trial, the LiDAR successfully measured the water level. Since the river surface was calm during the test, this success is attributed to the water's turbidity rather than surface rugosity. Suspended particles in turbid water enhance surface reflectivity for near-infrared wavelengths (905 nm), which aligns with the findings of Kupper, Sausen, and Campos [4].

Despite this success, the LiDAR readings in the river showed noticeable noise. This noise comes from many factors. The LiDAR node runs simple, self-written firmware with no averaging or filtering. The river test occurred under direct sunlight. As Kupper, Sausen, and Campos [4] note, 905 nm LiDAR sensors are highly sensitive to solar infrared light, which reduces their overall stability. Also, the LiDAR's narrow spot-beam captures the instant surface noise rather than a smoothed reading.

Testing the sensors at different tilt angles revealed further limitations. The radar sensor showed a sudden and sharp failure at a 20° tilt (−66.51% error), rather than a slow decline. This sharp drop points to a built-in limit in its internal software when pushed past its 15° rating.

6 Limitations and Future Work

All field trials were conducted under stable weather over a short period; storm or heavy-rain conditions were not tested. *Next step:* repeat the river deployment across multiple weather regimes.

The sensors were not tested under turbulent flow conditions, since the Timavo river was unusually calm during the trial, consistent with the severe drought affecting major European rivers in summer 2026. *Next step:* repeat the trial during higher-flow or post-rainfall conditions, when accurate readings matter most for flood warning.

The ultrasonic and radar sensors are commercial units with built-in processing, while the LiDAR node runs simple, custom-written firmware with no averaging or filtering. This makes the comparison not entirely fair, since the internal processing used by the commercial sensors is unknown.

Battery and solar performance under the adaptive transmission scheme were not evaluated. *Next step*: run a long-term deployment.

7 Conclusion

This project successfully developed and evaluated a river water level monitoring system using low-cost non-contact sensors. The evaluation demonstrates that the ideal sensor choice depends entirely on matching the device's strengths to the physical and environmental demands of the site. The radar sensor is highly stable but more costly, and it drops suddenly when tilted past its limits. The ultrasonic sensor is the cheapest but needs open space because of its wide beam. The LiDAR sensor reaches the furthest but is the most sensitive to nature, requiring careful management of sunlight, water clarity, and surface noise.

Whichever sensor is chosen, its readings are only useful if they reach decision-makers in time, which is where pairing LoRaWAN with a custom adaptive transmission strategy proves valuable. By automatically changing the sending rate based on how fast the water level changes, this system performs much better than the fixed-time methods seen in previous studies, such as the 5-minute fixed rate by Kabi et al. [2] or the 0.2 Hz rate by Martinez-Osuna et al. [3], providing fast, detailed data during urgent flood events while saving battery life during normal flow. Ultimately, this complete setup ensures that important, real-time data reaches decision-makers quickly, even from remote areas lacking cellular coverage.

References

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